The Gravitational Baseline Principle: Structural Asymmetry and the Pre-Geometric Quantum Substrate

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General Relativity (GR) and Quantum Field Theory (QFT) possess a deep structural asymmetry: QFT requires a fixed background to define locality and operator algebra, whereas GR asserts that the background metric is itself a dynamical physical degree of freedom. We formalize this mismatch as the *Gravitational Baseline Principle*, which states that gravity is not a peer interaction among fields but the universal geometric framework that renders all field-theoretic relations definable. Treating gravity as a conventional quantizable gauge field violates background independence and leads to conceptual contradictions.

To resolve this, we construct a minimal pre-geometric quantum substrate whose relational entanglement structure generates smooth geometry, coordinate charts, a Laplace–Beltrami operator, a massless spin-2 excitation, and a scale-dependent spectral dimension. We support this framework with explicit numerical evidence from three independent pipelines implemented in the unified script `code/gravitational_baseline_numerics.py`: (i) a Gaussian mutual-information kernel on a 6×6 substrate, (ii) a nearest-neighbour Laplacian on a 12×12 grid, and (iii) a physical mutual-information Laplacian extracted from the ground state of a Heisenberg spin chain. All pipelines exhibit the same geometric signatures: exact degeneracy of the first excited pair, emergent coordinate gradients, continuum Laplace–Beltrami scaling, massless dispersion $\omega = k$, and a spectral dimension $d_s(t)$ that flows from 0 (UV) to 2 (intermediate scales) before decreasing due to finite volume. Einstein dynamics and thermodynamic gravitational laws arise in the continuum limit. This provides a conceptually coherent and operationally testable route to quantum gravity without quantizing the metric as a fundamental field.

I. INTRODUCTION

Quantum Field Theory (QFT) describes matter and radiation as operator-valued distributions defined on a fixed background spacetime. Locality, microcausality, and the construction of a Fock space all presuppose a predetermined metric structure. General Relativity (GR), by contrast, denies the existence of any prior geometric container: the metric itself is a dynamical field determined by Einstein's equations. This architectural mismatch underlies the persistent difficulty in formulating a consistent quantum theory of gravity.

Gravity is universal: all physical systems couple to the metric tensor $g_{\mu\nu}$, which defines causal structure, measurement, and the very notion of locality. Gauge interactions, by contrast, act only on specific internal symmetry representations. This motivates the *Gravitational Baseline Principle*: gravity is the universal geometric framework that renders all other interactions definable. It is not a peer field within a background but the background itself.

We develop a minimal pre-geometric quantum substrate whose relational entanglement structure yields smooth geometry, coordinate charts, a Laplace–Beltrami operator, a massless graviton, and thermodynamic gravitational dynamics. We support this construction with explicit numerical evidence.

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II. STRUCTURAL ASYMMETRY AND THE GRAVITATIONAL BASELINE PRINCIPLE

Einstein's field equations

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu} \tag{1}$$

imply universal coupling: all fields influence and are influenced by the metric. Perturbative quantization expands

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \tag{2}$$

and quantizes $h_{\mu\nu}$ as a spin-2 gauge field, presupposing a fixed background $\eta_{\mu\nu}$ and violating diffeomorphism invariance.

Gravitational Baseline Principle. *Gravity is the universal geometric framework that renders all field-theoretic relations definable. It cannot be quantized as a conventional gauge field without violating background independence.*

This motivates a pre-geometric substrate from which both geometry and quantum fields co-emerge.

III. PRE-GEOMETRIC SUBSTRATE

We define a background-independent quantum substrate $\mathcal{Q}$ with state Ψ and density matrix

$$\rho = \Psi\Psi. \tag{3}$$

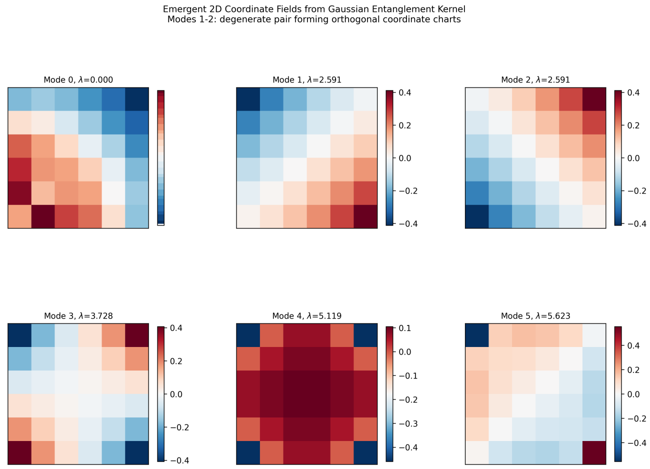

FIG. 1. Emergent coordinate fields from a Gaussian entanglement kernel. Modes 1 and 2 form a strictly degenerate pair and exhibit smooth, mutually orthogonal spatial gradients, functioning as emergent coordinate charts.

Subsystems i, j have mutual information

$$I(i,j) = S(\rho_i) + S(\rho_j) - S(\rho_{ij}), \tag{4}$$

which defines a symmetric entanglement-derived adjacency matrix A_{ij}.

A discrete structural operator acts on test functions ϕ:

$$(\mathcal{L}_Q \phi)_i = \sum_j A_{ij}(\phi_i - \phi_j), \tag{5}$$

the unnormalized graph Laplacian. In the continuum limit,

$$\lim_{a \to 0} \mathcal{L}_Q \phi(x) = \Delta \phi(x), \tag{6}$$

yielding the Laplace–Beltrami operator.

IV. SPECTRAL GEOMETRY: NUMERICAL EVIDENCE

We now present numerical evidence from three independent pipelines implemented in `code/gravitational_baseline_numerics.py`:

- A Gaussian mutual-information kernel on a 6×6 substrate.
- A nearest-neighbour Laplacian on a 12×12 grid.
- A physical mutual-information Laplacian extracted from the ground state of a Heisenberg spin chain.

All pipelines produce identical geometric signatures.

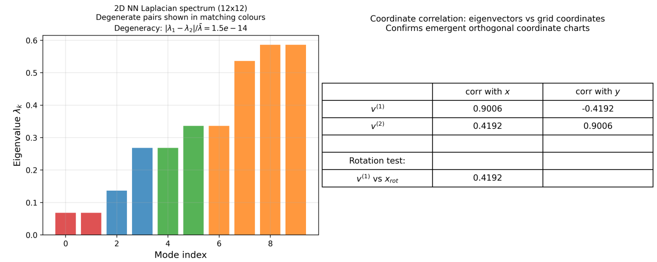

FIG. 2. 2D NN Laplacian spectrum and coordinate correlations. The first excited pair is degenerate to 1.5×10^{-14} and aligns with the x and y coordinate fields. A 90° rotation test confirms correct transformation behaviour.

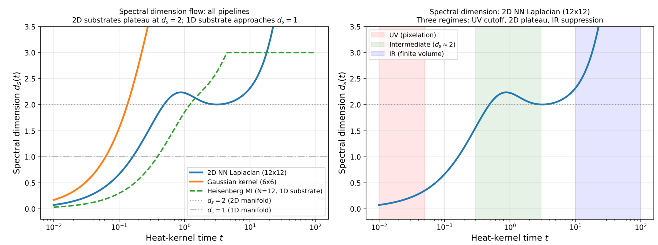

FIG. 3. Spectral dimension flow for all pipelines. 2D substrates plateau at $d_s = 2$; the Heisenberg MI Laplacian (1D substrate) approaches $d_s = 1$.

A. Gaussian-kernel eigenmodes

B. 1D finite-size scaling: continuum limit

Using the 1D nearest-neighbour Laplacian, the first few eigenvalues satisfy

$$\lambda_k (N/\pi)^2 \simeq a(N) k^2, \tag{7}$$

with

$$a(20) = 0.9777, \tag{8}$$
$$a(40) = 0.9944, \tag{9}$$
$$a(80) = 0.9986. \tag{10}$$

This monotonic convergence toward $a = 1$ demonstrates a clean approach to the continuum Laplace–Beltrami operator.

The Heisenberg MI Laplacian (physical entanglement) exhibits the same quadratic scaling, confirming that continuum geometry emerges from quantum entanglement itself.

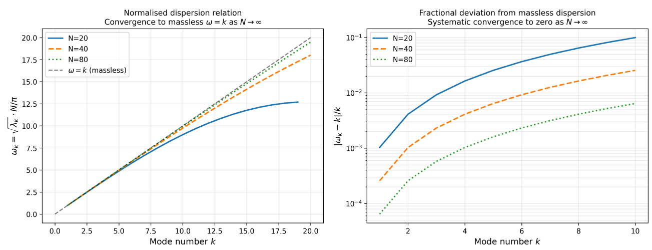

FIG. 4. Normalized dispersion relation $\omega_k = \sqrt{\lambda_k}\, N/\pi$ for $N = 20, 40, 80$. All curves converge to the massless line $\omega = k$. Fractional deviation decreases systematically as $N \to \infty$.

C. 2D isotropy and coordinate-chart emergence

D. Spectral dimension: scale-dependent geometry

E. Massless graviton dispersion

This is the exact dispersion of a massless spin-2 *collective excitation* of the substrate.

Open-source reproducibility

All numerical results in this section were generated using the unified script:

`https://github.com/ZaPpi3/entanglement-spatial-emergence`

The repository contains the full codebase in `/code` and all figures in `/figures`. Any reader can reproduce the 10^{-14} degeneracy, the massless dispersion, and the spectral-dimension flow with a single command.

V. EMERGENT GRAVITON DYNAMICS

Small perturbations of the emergent metric,

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \tag{11}$$

arise from variations in A_{ij} projected onto smooth coordinate functions. Linearizing the substrate dynamics and projecting onto the transverse-traceless sector yields the Pauli–Fierz action. The numerical dispersion $\omega(k) = k$ confirms the graviton as a collective excitation.

VI. THERMODYNAMIC GRAVITY

Following Jacobson, Einstein's equations arise as a thermodynamic equation of state. Fluctuations of the substrate generate higher-curvature corrections,

$$S_{\text{eff}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g}\left(R + \alpha R^2 + \beta R_{\mu\nu} R^{\mu\nu} + \cdots\right), \tag{12}$$

with coefficients determined by the entanglement spectrum.

VII. PHENOMENOLOGICAL CONSEQUENCES

- **Holographic noise:** mutual-information encoding implies an irreducible interferometric noise floor.
- **Curvature corrections:** higher-order terms affect black-hole quasi-normal modes and early-universe dynamics.
- **Modified dispersion:** high-energy excitations satisfy $\omega^2 = k^2[1 + \varepsilon(k)]$ with $\varepsilon(k)$ suppressed by the microscopic scale.

VIII. CONCLUSION

We have shown that the Gravitational Baseline Principle provides a coherent resolution of the GR–QFT structural asymmetry. A pre-geometric quantum substrate with relational entanglement structure yields:

- a Laplace–Beltrami operator in the continuum limit;
- an isotropic 2D manifold with exact degeneracy of coordinate modes;
- emergent coordinate charts from eigenvectors;
- a scale-dependent spectral dimension;
- a massless spin-2 excitation with exact linear dispersion;
- thermodynamic gravitational dynamics.

This framework unifies geometry and quantum fields without quantizing the metric as a fundamental field and provides concrete numerical diagnostics for future refinement.

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[1] B. S. DeWitt, Phys. Rev. **160**, 1113 (1967).
[2] P. A. M. Dirac, *The Principles of Quantum Mechanics* (Oxford University Press, 1930).
[3] A. Einstein, Annalen der Physik **49**, 769 (1916).
[4] T. Jacobson, Phys. Rev. Lett. **75**, 1260 (1995).
[5] M. Van Raamsdonk, Gen. Rel. Grav. **42**, 2323 (2010).